CURRICULUM VITAE

FELIX F. LOEFFLER

Address	Max Planck Institute of Colloids and Interfaces Department of Biomolecular Systems Am Muehlenberg 1 14476 Potsdam, Germany
Phone	+49 331 567-9359
Email	felix.loeffler@mpikg.mpg.de
Web	http://www.mpikg.mpg.de/SAT

PROFESSION

01/2017 – present **Group Leader**
Synthetic Array Technologies
Max Planck Institute of Colloids and Interfaces, Potsdam

05/2021 – 09/2021 **Parental leave**

07/2014 – 12/2016 **Young Investigator & Project Leader**
Microarray Technology and Application
Karlsruhe Institute of Technology, Institute of Microstructure Technology

01/2014 – 06/2014 **Visiting Scholar**
Molecular Virology and Immunology of Dengue Virus
University of California, Berkeley, USA

05/2012 – 12/2013 **Postdoctoral Fellow**
Microstructuring for Combinatorial Chemistry
Karlsruhe Institute of Technology, Institute of Microstructure Technology

Peptide Microarrays for Disease Research
German Cancer Research Center (DKFZ), Heidelberg

EDUCATION

10/2009 – 04/2012 (30 months) **PhD (summa cum laude, highest honors)**
Selective aerosol particle deposition on CMOS electrode chips
University of Heidelberg, Kirchhoff Institute of Physics &
German Cancer Research Center (DKFZ), Heidelberg

10/2004 – 09/2009 **Physics Diploma (grade 1.3)** and certificate in biophysics
University of Heidelberg

INDUSTRY EXPERIENCE

Since 2019 Consultant for array, diagnostics, and printing technologies (Third Bridge, Goldfish Consulting)

01/2015 – 12/2016 Statistical data analysis (PEPperPRINT GmbH)

SCIENTIFIC INTERESTS AND EXPERTISE

- **Biotechnology & Bioengineering:** Microarrays, Infectious Diseases, Immunology, "Omics" Research, Glycobiology, Phage Display, Molecular Diagnostics & Interactions, Lab-on-a-Chip, Biosensors, High-Throughput Screening, Statistical Data Analysis
- **Process & Chemical Engineering, Chemistry:** Surface Engineering, Parallel Synthesis Technology, Glycochemistry, Polymer & Physical Chemistry, Thermal Processing, Solid Phase Synthesis (Peptides, Glycans, Oligonucleotides), Biochemistry, Photoelectrochemistry
- **Engineering, Physics & Technology:** Additive Manufacturing (FDM, SLM, Lithography), Micro- & Nanoprinting, Laser-Based Material Transfer, Surface Science, Microarray Technologies, 3D Design, Programming (C, MATLAB, Python), Machine Learning, Data Analysis, Laser Processing, Image Analysis, Particle and Polymer Patterning, Optics, Numerical Simulations, Fluid and Aerosol Dynamics
- **Materials Science:** Nanomaterials Synthesis, Anti-Counterfeiting, Graphitic Carbon Nitrides, Sustainable Chemistry, Photocatalysts

TEACHING, SUPERVISION & MANAGEMENT

Teaching

SS2023 & SS2021 SS2020	**Technical Chemistry**, 2 SWS, Chemistry Department, *Free University Berlin, Germany* Thermodynamics, Flow Chemistry, Separation Techniques, Sustainable Polymers & Chemistry, Oil Refining & Petrochemistry
SS2019 & WS2019/20	**High-Throughput Technologies – Microarrays**, 2 SWS, *Bioprocess Engineering, Technical University of Berlin, Germany* High-Throughput Biological/Chemical Technologies, Array and Printing Technologies, Solid Phase Synthesis, Automated Synthesis, Molecular Interactions, Microarray Applications, Statistical Data Analysis
SS2015	**Physics for engineers** (optics lectures), *Karlsruhe Institute of Technology, Germany*

Supervision

Since 2017	**5 Postdocs** (4 finished, 1 ongoing), **7 PhD students** (4 finished, 3 ongoing), **3 MSc students** (3 finished), **2 visiting PhD students** (1 finished - 12 months, 1 ongoing - 7 months), 5 research assistants at Master's level (5 finished)
2010 – 2016	2 PhD students (Physicist, Bioengineer), 1 MSc & 1 BSc student

Management

Since 2017	Departmental seminar co-organization (Biomolecular Systems Department), IT committee, library committee, working hour documentation committee, Max-Planck-Institute of Colloids and Interfaces, Potsdam, Germany
2016	Co-founder, startup company ("Stochastic combinatorics" GbR), Karlsruhe, Germany
2014 – 2016	Organizer of the Seminar "Peptide Arrays and Antibody Libraries", Karlsruhe Institute of Technology, Germany
2014 – 2016	Member of the Young Investigator Network (YIN), Public relations team

AWARDS, PRIZES & HONORS

10/2021 **Carl-Zeiss-Foundation selection committee member**, Nexus Program 2022 – 2024

10/2020 **Early Career Advisory Board**, *Chemistry – A European Journal*

07/2019 **ERC Starting Grant 2019** reserve list candidate (rated "A"), not funded due to budgetary cutoff

03/2017 **NanoMatFutur competition (2.25 m€)**, **1st place**, German Federal Ministry of Education and Research (BMBF)

10/2014 **Young Academics Award (1st place)**, Gips-Schuele-Foundation (10,000 €)

05/2014 Selected for a presentation in the master class of Prof. Rolf Zinkernagel at the 64th Lindau Nobel Laureate Meeting, Lindau/Germany

03/2014 Selected for the 64th **Lindau Nobel Laureate Meeting**, Lindau/Germany

11/2013 Selected for the Global Young Scientists Summit 2014, Singapore

11/2013 Foreign exchange scholarship to Berkeley/USA by the Karlsruhe House of Young Scientists (6,500 €)

11/2013 1st place poster presentation, scholars meeting of the Carl-Zeiss-Foundation, Oberkochen/Germany

09/2013 Carl Zeiss Foundation postdoctoral research scholarship **(200,000 €)**

11/2013 **Best presentation award**, International Workshop "Adaptive optical systems for particle-based combinatorial chemistry", KIT, 2011, Karlsruhe/Germany

06/2010 Poster presentation award, 1st place, German Cancer Research Center PhD Retreat, Weil der Stadt/Germany

04/2010 ForSys travel stipend to "Systems biology of human disease 2010", Boston/USA (1,500 €)

10/2009 **PhD stipend** (3 years) of the German Cancer Research Center

INVITED PRESENTATIONS & ORGANIZATION OF EVENTS

06/2024 International Symposium on Laser Precision Microfabrication, San Sebastián, Spain, 11-14 June 2024

11/2023 Karlsruhe Institute of Technology, Institute of Microstructure Technology, Eggenstein-Leopoldshafen, Germany, 20 November 2023

06/2023 University of Heidelberg, Flagship Initiative Engineering Molecular Systems, Heidelberg, Germany, 19 June 2023

03/2023 University of Amsterdam, Van 't Hoff Institute for Molecular Sciences, Amsterdam, The Netherlands, 27 March 2023

02/2023 Cambridge University & Max Planck Institute of Colloids and Interfaces Joint Workshop, Prof. Silvia Vignolini, Potsdam, Germany, 6 Feb 2023

02/2023 Max Planck Institute for Solid State Research, Department of Prof. B. Lotsch, Stuttgart, Germany, 3 Feb 2023

02/2023 University of Stuttgart & Fraunhofer Institute for Interfacial Engineering and Biotechnology (IGB), Stuttgart, Germany, 2 Feb 2023

11/2022 Nanobiotechnology Series (online), School of Engineering and Sciences, Tecnológico de Monterrey, Monterrey, Mexico, 2 Nov 2022

05/2022 Institute of Biological and Chemical Systems, Karlsruhe Institute of Technology, Karlsruhe, Germany, 6 May 2022

03/2022 Early Career Advisory Board of *Chemistry – A European Journal*, virtual event, 31 Mar 2022

04/2021 **Session chair**, Healthcare and Life Science & Entrepreneurship, *online event*, Technical University of Denmark, 29 – 30 Apr 2021

02/2021 **Keynote speaker**, ProMatLeben Conference (online), 2 – 3 Feb 2021

05/2020 DNA Nanotechnology 2020, Jena, Germany, 7 – 9 May 2020, *cancelled due to corona situation*

09/2019 **Organizer**, Ringberg 2019 Conference, Germany, "Photocatalysis and Synthetic Array Technologies", 2 – 6 Sep 2019

05/2019 BioCHIP Berlin 2019 Conference, Germany 7 – 8 May 2019

02/2019 Harris Lab, University of California, Berkeley, USA, 8 Feb 2019

02/2019 **Keynote speaker & session chair**, SPIE Photonics West, "Microfluidics, BioMEMS, and Medical Microsystems XVII", San Francisco, USA, 2 – 7 Feb 2019

01/2019 Max-Planck – Nanyang Technical University Joint Lab of Artificial Senses, Singapore, 13 – 15 January 2019

12/2018 **Organizer**, Biomolecular Systems Day 2018, Potsdam, Germany

01/2018 Universität Potsdam, Institute of Chemistry, Potsdam, Germany, 29 Jan 2018

09/2017 GDCh – Fachgruppe Makromolekulare Chemie – Workshop ("Habilitanden-Workshop"), BASF, Ludwigshafen, 26 Sep 2017

05/2017 Universität Potsdam, Institute of Physics and Astronomy, Potsdam, Germany, 5 May 2017

05/2016 8th Indo-German Frontiers of Engineering Symposium 2016 by the Humboldt Foundation, Potsdam, Germany, 19 – 22 May 2016

07/2014 64th Lindau Nobel Laureate Meeting, Lindau, Germany, 28 Jun – 4 Jul 2014 (**invited presentation as a young scientist, Prof. R. Zinkernagel**)

11/2011 International Workshop "Adaptive optical systems for particle-based combinatorial chemistry", KIT, Karlsruhe, Germany, 16 – 18 Nov 2011

LEADERSHIP TRAINING

Since 2017 Max Planck Society, Leadership Education Program

2015 – 2016 Young Investigator Network (YIN, KIT) Continuing Education Program

>15 total days of individual leadership training

COLLABORATIONS

National

MPICI Potsdam: Prof. Peter Seeberger, Prof. Markus Antonietti, Prof. Silvia Vignolini, Dr. Martina Delbianco, Dr. Lukas Zeininger, Dr. Thomas Parton, Dr. Shahrouz Amini; *KIT Karlsruhe:* Prof. Stefan Bräse, Prof. Frank Breitling, Prof. Jan Korvink, Dr. Dario Mager, Prof. Alexander Nesterov-Müller; *UKE Hamburg:* Prof. Marylyn Addo; *Uni Marburg:* Prof. Stefan Becker; *Uni Braunschweig:* Prof. Stefan Dübel, Prof. Michael Hust; *Uni Köln:* Prof. Florian Klein; *TiHo Hannover:* Prof. Bernd Lepenies; *UK Frankfurt:* Dr. Timo Wolf; *Uni Potsdam:* Prof. Svetlana Santer; *FAU Erlangen:* Prof. Karl Mandel

International

Netherlands: Prof. Jon Laman, UMH Groningen; Prof. Francesco Mutti, Prof. Hong Zhang, University of Amsterdam; *UK:* Prof. Molly Stevens, University of Oxford; Dr. Bernhard Schmidt, University of Glasgow; *China:* Prof. Oleksandr Savateev, CU Hong Kong; Prof. Xingjun Zhu, ShanghaiTech University; *Austria:* Prof. Christoph Rademacher, Uni Vienna; *Canada:* Prof. Nabyl Merbouh, SFU Vancouver; Prof. Natalie Strynadka, UBC Vancouver; *USA*: Prof. Thomas Jänisch, UC Denver; Prof. Ernesto Marquez, Uni Pittsburg; *Brazil:* Dr. Roberto Lins Fundação Oswaldo Cruz, Recife; *Norway:* Prof. Andreas Lossius, Uni Oslo

SOCIETIES, BOARD MEMBERSHIPS & REVIEWING

Societies

- German Society for Chemical Engineering and Biotechnology (DECHEMA)
- German Chemical Society (GDCh), Macromolecular Chemistry Section
- German Physical Society (DPG), Condensed Matter Section
- Verein zur Förderung der Forschung an biomolekularen Systemen e.V., *Treasurer*

Board & Committee member

- **Early Career Advisory Board**, *Chemistry – A European Journal*, since 2020
- **Selection committee, Carl Zeiss Foundation "Nexus" Program**, since 2022, selecting 8 young scientists per year (*1.5 m€ each*) for an independent research career

Reviewer for scientific proposals

Deutsche Forschungsgemeinschaft (DFG), Carl Zeiss Foundation, Alexander von Humboldt Foundation, United States-Israel Binational Science Foundation

Reviewer for scientific journals

Nature Nanotechnology, New England Journal of Medicine, Nature Communications, Advanced Materials, Advanced Functional Materials, Small, Small Methods, Accounts of Materials Research, JACS Au, Analytical Chemistry, Chemistry – A European Journal, ACS Infectious Diseases, Langmuir, Advanced Engineering Materials, Scientific Reports, and others

PUBLIC OUTREACH

- Several reports in the news on sustainable anti-counterfeiting (*e.g.*, Deutsche Apotheker Zeitung, Physik Journal, Technik in Bayern/VDI, MaxPlanckResearch, Frost & Sullivan report), 06–12/2023
- Bürgergespräch „Materialinnovationen für Gesundheit und Lebensqualität", Berlin, Germany, 8 Oct 2020
- MPICI Girl's Day 2019: "Robots, Lasers, & Chemistry" Laboratory presentation, 2019
- Radio comment on microarrays, Antenne Brandenburg, 2019
- Supervision of high school student interns 2019 – 2020

LANGUAGE SKILLS

- German (native)
- English (fluent)

PATENTS (authors in alphabetical order)

1. EP23173768.5: **F.F. Loeffler (shared main inventor)**, J. Zhang. Anti-counterfeiting labels with solid-state fluorescent carbon nanodots. Filed May 16, **2023**
2. EP22151236.1: **F.F. Loeffler (shared main inventor)**, A. Savateev, J. Zhang, Y. Zhu. Novel carbon nitride compositions and their use as photoelectrodes. Filed January 12, **2022**
3. EP18213784.4: S. Eickelmann, J. Heidepriem, **F.F. Loeffler (main inventor)**, M. Mende, G. Paris, P. H. Seeberger, A. Tsouka. Method and device for producing saccharides and saccharide arrays. Filed December 18, **2018**
4. PCT/EP2018/075397: S. Dübel, V. Fuehner, R. Gerhard, M. Hust, **F.F. Loeffler**. Vaccine and antibody against Clostridiodes difficile toxin. Filed September 19, **2018**
5. EP17209651.3: N. Fischer, T. Jaenisch, **F.F. Loeffler (shared main inventor)**, T. Magalhaes, E.T.A. Marques. NS2b as marker for Zika virus infections. Filed December 21, **2017**
6. Patent application 102015117567.3: D. Althuon, F. Breitling, V. Bykovskaya, **F.F. Loeffler**, A. Nesterov-Mueller, R. Popov, B. Ridder, C. v. Bojnicic-Kninski. Ultra-hochdichte Oligomerarrays und Verfahren zu deren Herstellung. Filed October 15, 2015; DE10 2015 117 567, issued January 18, **2017**

7. PCT/EP2014/001046*: F. Breitling, V. Bykovskaya, K. Leibe, **F.F. Loeffler (shared main inventor)**, F. Maerkle, A. Nesterov-Mueller, S. Schillo, C. v. Bojnicic-Kninski. Method for combinatorial particle manipulation for producing high-density molecular arrays, particularly peptide arrays, and molecular arrays which can be obtained therefrom. Filed April 17, **2014**, issued **2019**
8. DE102013113169A1: F. Breitling, H.-A. Freudigmann, **F.F. Loeffler**, K.L. Hahn, F. Maerkle, B. Muenster, A. Nesterov-Mueller, S. Schillo, J. Striffler, T. Rupp. „Vorrichtung und Verfahren zur Herstellung von Partikelschichten und deren Verwendung“. Filed November 28, **2013**
9. PCT/EP2013/001141*: F. Breitling, V. Bykovskaya, K. Leibe, **F.F. Loeffler (shared main inventor)**, F. Maerkle, A. Nesterov-Mueller, S. Schillo, C. v. Bojnicic-Kninski. Method for combinatorial particle manipulation for producing high-density molecular arrays, particularly peptide arrays, and molecular arrays which can be obtained therefrom. Filed April 17, **2013**

* application was amended and filed as EP2013/001141 and EP2014/001046

PUBLICATIONS

Top 5 selected publications

- J. Zhang, Y. Liu, C. Njel, S. Ronneberger, N. V. Tarakina, **F.F. Loeffler***. An all-in-one nanoprinting approach for the synthesis of a nanofilm library for unclonable anticounterfeiting applications. *Nature Nanotechnology*, **2023**, 18, 1027–1035
https://doi.org/10.1038/s41565-023-01405-3 ***Impact factor: 38.3***
- J. Zhang, R. Tan, Y. Liu, M. Albino, W. Zhang, M.M. Stevens, **F.F. Loeffler***. Printed smart devices for anti-counterfeiting allowing precise authentication with household equipment. *Nature Communications*, **2024**, 15, 1040
https://doi.org/10.1038/s41467-024-45428-3 ***Impact factor: 16.6***
- A. Tsouka, P. Dallabernardina, M. Mende, E.T. Sletten, S. Leichnitz, K. Bienert, K.L.M. Hoang, P.H. Seeberger, **F.F. Loeffler***. VaporSPOT: Parallel synthesis of oligosaccharides on membranes. *Journal of the American Chemical Society,* **2022**, 144, 19832–19837
https://doi.org/10.1021/jacs.2c07285 ***Impact factor: 15.0***
- G. Paris, J. Heidepriem, A. Tsouka, Y. Liu, D. S. Mattes, S. Pinzón Martín, P. Dallabernardina, M. Mende, C. Lindner, R. Wawrzinek, C. Rademacher, P.H. Seeberger, F. Breitling, F.R. Bischoff, T. Wolf, **F.F. Loeffler***. Automated laser-transfer synthesis of high-density microarrays for infectious disease screening. *Advanced Materials*, **2022**, 34, 2200359
https://doi.org/10.1002/adma.202200359 ***Impact factor: 29.4***
- J. Zhang, Y. Zou, T. Heil, C. Njel, S. Eickelmann, S. Ronneberger, V. Strauss, P.H. Seeberger, A. Savateev, **F.F. Loeffler***. Laser-driven growth of structurally defined transition metal oxide nanocrystals on carbon nitride photoelectrodes in milliseconds. *Nature Communications*, **2021**, 12, 3224
https://doi.org/10.1038/s41467-021-23367-7 ***Impact factor: 16.6***

Peer-reviewed publications (total: 64, *corresponding author: 35, first author: 8)

IF	#	Reference
29.4	1.	Y. Liu, Z. Wei, J. Zhang, Y. Xu, J. Zhou, Z. Ma, F. G. Mutti, H. Zhang*, X. Zhu*, **F.F. Loeffler***. Customized enhancement of thermal sensitivity of tumors at different subcutaneous depths by multichannel lanthanide nanocomposites. *Advanced Materials, accepted*
20.6	2.	Y. Liu*, Z. Wei, F.G. Mutti, H. Zhang, **F.F. Loeffler***. Redox-responsive inorganic fluorescent nanoprobes for serodiagnosis and bioimaging. *Coordination Chemistry Reviews, accepted*
19.0	3.	A. Tsouka, M. Mende, J. Heidepriem, P. Dallabernardina, K. Bienert, P.H. Seeberger, **F.F. Loeffler***. VaporLIFT: On-chip chemical synthesis of glycan microarrays. *Advanced Functional Materials*, **2024**, 2310980
3.9	4.	R. Tan, Y. Liu, Y. Tu*, **F.F. Loeffler***. Laser-Induced Phase Engineering of MoS2 Nanofilm for Enhanced Photoelectrochemical Performance. *RSC Advances*, **2024**, 14, 4730–4733
16.6	5.	J. Zhang, R. Tan, Y. Liu, M. Albino, W. Zhang, M.M. Stevens, **F.F. Loeffler***. Printed smart devices for anti-counterfeiting allowing precise authentication with household equipment. *Nature Communications*, **2024**, 15, 1040
9.2	6.	L. Mayer, L.M. Weskamm, A. Fathi, M. Kono, J. Heidepriem, V. Krähling, S. Mellinghoff, M.L. Ly, M. Friedrich, S. Hardtke, S. Borregaard, T. Hesterkamp, **F.F. Loeffler**, A. Volz, G. Sutter, S. Becker, C. Dahlke, M.M. Addo. MVA-based vaccine candidates encoding the native or prefusion-stabilized SARS-CoV-2 spike reveal differential immunogenicity in humans. *npj Vaccines,* **2024**, 9, 20
16.6	7.	J. Zhang, Y. Zhu, C. Njel, Y. Liu, P. Dallabernardina, M.M. Stevens, P.H. Seeberger, A. Savateev*, **F.F. Loeffler***. Metal-free photoanodes for C-H functionalization. *Nature Communications*, **2023**, 14, 7104
29.4	8.	Y. Liu, Z. Wei, M. Damian, X. Zhu, T. Knaus, H. Zhang*, F.G. Mutti*, **F.F. Loeffler***. A recyclable and robust optical nanoprobe with engineered enzymes for sustainable serodiagnostics. *Advanced Materials,* **2023**, 35, 2306615
38.3	9.	J. Zhang, Y. Liu, C. Njel, S. Ronneberger, N. V. Tarakina, **F.F. Loeffler***. An all-in-one nanoprinting approach for the synthesis of a nanofilm library for unclonable anticounterfeiting applications. *Nature Nanotechnology,* **2023**, 18, 1027–1035
16.6	10.	Y. Liu, X. Zhu, Z. Wei, K. Wu, J. Zhang, F.G. Mutti, H. Zhang*, **F.F. Loeffler***, J. Zhou*. Multi-Channel Lanthanide Nanocomposites for Customized Synergistic Treatment of Orthotopic Multi-Tumor Cases. *Angewandte Chemie International Edition,* **2023**, 62, e202303570
19.0	11.	S. Ronneberger, J. Zhang, Y. Liu, **F.F. Loeffler***. Solid ink laser patterning for high-resolution information labels with supervised learning readout. *Advanced Functional Materials*, **2023**, 33, 2210116
15.0	12.	A. Tsouka, P. Dallabernardina, M. Mende, E.T. Sletten, S. Leichnitz, K. Bienert, K.L.M. Hoang, P.H. Seeberger, **F.F. Loeffler***. VaporSPOT: Parallel synthesis of oligosaccharides on membranes. *Journal of the American Chemical Society,* **2022**, 144, 19832–19837
16.6	13.	A. Fathi, C. Dahlke, A. Kupke, V. Krähling, N. M. A. Okba, M. Raadsen, J. Heidepriem, G. Paris, M. Müller, S. Lassen, M. Kluever, S. Halwe, C. Rohde, M. Eickmann, A. Volz, T. Koch, M. L. Ly, M. Friedrich, R. Fux, A. Tscherne, G. Kalodimou, S. Schmiedel, A. W. Lohse, V. Corman, T. Hesterkamp, C. Drosten, **F.F. Loeffler**, B. L. Haagmans, G. Sutter, S. Becker, M.M. Addo. Increased neutralization and IgG epitope identification after MVA-MERS-S booster vaccination against Middle East respiratory syndrome. *Nature Communications,* **2022**, 13, 4182
29.4	14.	G. Paris, J. Heidepriem, A. Tsouka, Y. Liu, D. S. Mattes, S. Pinzón Martín, P. Dallabernardina, M. Mende, C. Lindner, R. Wawrzinek, C. Rademacher, P.H. Seeberger, F. Breitling, F.R. Bischoff, T. Wolf, **F.F. Loeffler***. Automated laser-transfer synthesis of high-density microarrays for infectious disease screening. *Advanced Materials*, **2022**, 34, 2200359
3.9	15.	S. Eickelmann, S. Moon, Y. Liu, B. Bitterer, S. Ronneberger, D. Bierbaum, F. Breitling, **F.F. Loeffler***. Assessing Polymer-Surface Adhesion with a Polymer Collection. *Langmuir,* **2022**, 38, 2220–2226
2.7	16.	G. Paris, D. Bierbaum, M. Paris, D. Mager, **F.F. Loeffler***. Development and Experimental Assessment of a Model for the Material Deposition by Laser-Induced Forward Transfer. *Applied Sciences*, **2022**, 12, 1361

14.6 17. M. Hepp, H. Wang, K. Derr, S. Delacroix, S. Ronneberger, **F.F. Loeffler**, B. Butz, V. Strauss. Trained Laser-Patterned Carbon as High-Performance Mechanical Sensors. *npj Flexible Electronics,* **2022**, 6, 1–9

29.4 18. J. Zhang*, Y. Liu, S. Ronneberger, N.V. Tarakina, N. Merbouh, **F.F. Loeffler***. Nanolayer laser absorber for femtoliter chemistry in polymer reactors. *Advanced Materials*, **2022**, 34, 2108493

3.2 19. P. Dallabernardina, V. Benazzi, J.D. Laman, P.H. Seeberger, **F.F. Loeffler***. Automated Glycan Assembly of Peptidoglycan Backbone Fragments. *Organic & Biomolecular Chemistry,* **2021**, 19, 9829–9832

4.3 20. Y. Liu, P.H. Seeberger, N. Merbouh*, **F.F. Loeffler***. Position Matters: Fluorescent Positional Isomers for Reversible Multichannel Encryption Devices. *Chemistry – A European Journal,* **2021**, 27, 16098–16102

5.5 21. A. Tsouka, K. Hoetzel, M. Mende, J. Heidepriem, G. Paris, S. Eickelmann, P.H. Seeberger, B. Lepenies, **F.F. Loeffler***. Probing multivalent carbohydrate-protein interactions with on-chip synthesized glycopeptides using different functionalized surfaces. *Frontiers in Chemistry,* **2021**, 9, 766932

4.1 22. **F.F. Loeffler**, I.F.T. Viana, N. Fischer, D.F. Coêlho, C.S. Silva, C.M.C.S. Araújo, B.H.S. Leite, R. Durães-Carvalho, T. Magalhães, C.N.L. Morais, M.T. Cordeiro, R.D. Lins, E.T.A. Marques, T. Jaenisch. Identification of a Zika NS2B epitope as a biomarker for severe clinical phenotypes. *RSC Medicinal Chemistry,* **2021**, 12, 1525–1539

3.7 23. J. Heidepriem, C. Dahlke, R. Kobbe, R. Santer, T. Koch, A. Fathi, B.M.S. Seco, M.L. Ly, S. Schmiedel, D. Schwinge, S. Serna, K. Sellrie, N.-C. Reichardt, ID-UKE COVID-19 study group, P.H. Seeberger, M.M. Addo*, **F.F. Loeffler***. Longitudinal development of antibody responses in COVID-19 patients of different severity with ELISA, peptide, and glycan arrays: an immunological case series. *Pathogens,* **2021**, 10, 438

16.6 24. J. Zhang, Y. Zou, S. Eickelmann, C. Njel, T. Heil, S. Ronneberger, V. Strauss, P.H. Seeberger, A. Savateev, **F.F. Loeffler***. Laser-driven growth of structurally defined transition metal oxide nanocrystals on carbon nitride photoelectrodes in milliseconds. *Nature Communications*, **2021**, 12, 3224

5.4 25. S. Eickelmann*, S. Ronneberger, J. Zhang, G. Paris, **F.F. Loeffler***. Alkanes as intelligent surface thermometers: a facile approach to characterize short-lived temperature gradients on the micrometer scale. *Advanced Materials Interfaces,* **2021**, 8, 2001626

7.1 26. I.K. Ilic, A. Tsouka, M. Perovic, J. Hwang, T. Heil, **F.F. Loeffler**, M. Oschatz, M. Antonietti, C. Liedel. Sustainable Cathodes for Lithium-Ion Energy Storage Devices Based on Tannic Acid – Towards Eco-Friendly Energy Storage. *Advanced Sustainable Systems*, **2021**, 5, 2000206

4.7 27. S. Delacroix, A. Zieleniewska, A.J. Ferguson, J.L. Blackburn, S. Ronneberger, **F.F. Loeffler**, V. Strauss. Using Carbon Laser-Patterning to Produce Flexible, Metal-Free Humidity Sensors. *ACS Applied Electronic Materials*, **2020**, 2, 4146–4154

4.4 28. K. Heiss*, J. Heidepriem, N. Fischer, L.K. Weber, C. Dahlke, T. Jaenisch, **F.F. Loeffler***. Rapid response to pandemic threats: immunogenic epitope detection of pandemic pathogens for diagnostics and vaccine development using peptide microarrays. *Journal of Proteome Research*, **2020**, 19, 4339–4354

4.3 29. M. Mende, A. Tsouka, J. Heidepriem, G. Paris, D.S. Mattes, S. Eickelmann, V. Bordoni, R. Wawrzinek, F.F. Fuchsberger, P.H. Seeberger, C. Rademacher, M. Delbianco, A. Mallagaray, **F.F. Loeffler***. On-Chip Neo-Glycopeptide Synthesis for Multivalent Glycan Presentation. *Chemistry – A European Journal,* **2020**, 26, 9954

4.7 30. J. Heidepriem, V. Krähling, C. Dahlke, T. Wolf, F. Klein, M.M. Addo*, S. Becker*, **F.F. Loeffler***. Epitopes of naturally acquired and vaccine-induced anti-Ebola virus glycoprotein antibodies in single amino acid resolution. *Biotechnology Journal*, **2020**, 15, 2000069

6.7 31. G. Paris, A. Klinkusch, J. Heidepriem, A. Tsouka, J. Zhang, M. Mende, D. Mattes, D. Mager, H. Riegler, S. Eickelmann*, **F.F. Loeffler***. Laser-induced forward transfer of soft material nanolayers with millisecond pulses shows contact-based material deposition. *Applied Surface Science*, **2020**, 508, 144973

4.3 32. J. Zhang, S. Gim, G. Paris, P. Dallabernardina, C. N. Z. Schmitt, S. Eickelmann, **F.F. Loeffler***. Ultrasonic-assisted synthesis of highly defined silver nanodimers by self-assembly for improved surface-enhanced Raman spectroscopy. *Chemistry – A European Journal*, **2020**, 26, 1243

6.8 33. S. Eickelmann*, A. Tsouka, J. Heidepriem, G. Paris, J. Zhang, V. Molinari, M. Mende, **F.F. Loeffler***. A Low-Cost Laser-Based Nano-3D Polymer Printer for Rapid Surface Patterning and Chemical Synthesis of Peptide and Glycan Microarrays. *Advanced Materials Technologies,* **2019***,* 4, 1900503

3.4 34. M. Mende, V. Bordoni, A. Tsouka, **F.F. Loeffler**, M. Delbianco, P. H. Seeberger. Multivalent glycan arrays. *Faraday Discussions*, **2019**, 219, 9

2.7 35. T. C. Foertsch, A. T. Davis, R. Popov, C. v. Bojničić-Kninski, F. E. Held, S. B. Tsogoeva, **F.F. Loeffler**, A. Nesterov-Mueller. Spatial modes of laser-induced mass transfer in microgaps. *Applied Sciences*, **2019**, 9, 1303

6.1 36. M. M. Aeinehvand, L. Weber, M. Jiménez Moreno, A. Palermo, M. Bauer, **F.F. Loeffler**, F. Breitling, F. Ibrahim, J. Korvink, M. Madou, D. Mager, S. Martinez-Chapa. Elastic reversible valves on centrifugal microfluidic platforms. *Lab on a Chip*, **2019**, 19, 1090–1100

7.0 37. T. Jaenisch, K. Heiss, N. Fischer, C. Geiger, F. R. Bischoff, G. Moldenhauer, L. Rychlewski, A. Sié, B. Coulibaly, P.H. Seeberger, L.S. Wyrwicz, F. Breitling, **F.F. Loeffler***. High-density peptide arrays help to identify linear immunogenic B cell epitopes in individuals naturally exposed to malaria infection. *Molecular & Cellular Proteomics*, **2019**, 18, 642–656

4.6 38. D.S. Mattes, B. Streit, D.R. Bhandari, J. Greifenstein, T.C. Foertsch, S. Muench, B. Ridder, C.v. Bojničić-Kninski, A. Nesterov-Mueller, B. Spengler, U. Schepers, S. Bräse, **F.F. Loeffler*** (**shared last/corresponding author**), F. Breitling*. Synthesis of peptoid arrays via matrix-based combinatorial laser transfer. *Macromolecular Rapid Communications*, **2018**, 40, 1800533

5.2 39. V. Fühner, P. Heine, S. Helmsing, S. Goy, J. Heidepriem, **F.F. Loeffler**, S. Dübel, R. Gerhard, M. Hust. Development of neutralizing and non-neutralizing antibodies targeting TcdB of Clostridium difficile. *Frontiers in Microbiology*, **2018**, 9, 2908

29.4 40. J. Atwater, D.S. Mattes, B. Streit, C.v. Bojničić-Kninski, **F.F. Loeffler**, F. Breitling, H. Fuchs, M. Hirtz. Combinatorial synthesis of macromolecular arrays by microchannel cantilever spotting (µCS). *Advanced Materials,* **2018**, 30, 1801632

12.4 41. D.S. Mattes, S. Rentschler, T.C. Foertsch, S.W. Münch, **F.F. Loeffler**, A. Nesterov-Mueller, S. Bräse. F. Breitling. A Trifunctional Linker for Purified Three Dimensionally Assembled Peptide Structure Arrays. *Small Methods,* **2018**, 2, 1700205

4.7 42. A. Palermo, L.K. Weber, S. Rentschler, A. Isse, M. Sedlmayr, K. Herbster, V. List, J. Hubbuch, **F.F. Loeffler***, A. Nesterov-Mueller*, F. Breitling*. Epitope mapping and fingerprint analysis of antibodies elicited by vaccinations using high density peptide arrays. *Biotechnology Journal*, **2017**, 12, 1700197

19.0 43. C.v. Bojničić-Kninski, R. Popov, E. Dörsam, **F.F. Loeffler**, F. Breitling, A. Nesterov-Mueller. Combinatorial Particle Patterning. *Advanced Functional Materials,* **2017**, 27, 1703511

4.1 44. M.C.L.C. Freire, L. Pol-Fachin, D.F. Coêlho, I.F.T. Viana, T. Magalhães, N. Fischer, **F.F. Loeffler**, T. Jaenisch, R.F. França, E.T.A. Marques, R.D. Lins. Mapping Putative B cell Zika Virus NS1 Epitopes Provides Molecular-Basis For Anti-NS1 Antibody Discrimination Between Zika And Dengue Viruses. *ACS Omega*, **2017**, 2, 3913–3920

2.7 45. L.K. Weber, A. Isse, S. Rentschler, R.E. Kneusel, A. Palermo, J. Hubbuch, A. Nesterov-Mueller*, F. Breitling*, **F.F. Loeffler***. Antibody fingerprints in Lyme disease deciphered with high density peptide arrays. *Engineering in Life Sciences*, **2017**, 17, 1078–1087

2.2 46. L.K. Weber, A. Palermo, J. Kügler, O. Armant, A. Isse, S. Rentschler, T. Jaenisch, J. Hubbuch, S. Duebel, A. Nesterov-Mueller*, F. Breitling*, **F.F. Loeffler***. Single amino acid fingerprinting of the human antibody repertoire with high density peptide arrays. *Journal of Immunological Methods*, **2017**, 443, 45–54

6.7 47. B. Ridder, T.C. Foertsch, A. Welle, D. Mattes, C. v. Bojnicic-Kninski, **F.F. Loeffler**, A. Nesterov-Mueller, M.A.R. Meier, F. Breitling. Development of a poly(dimethylacrylamide) based matrix material for solid phase high density peptide array synthesis employing a laser based material transfer. *Applied Surface Science*, **2016**, 389, 942–951

19.0 48. C. v. Bojnicic-Kninski, V. Bykovskaya, F. Maerkle, R. Popov, A. Palermo, D. S. Mattes, L. K. Weber, B. Ridder, T. C. Foertsch, A. Welle, **F.F. Loeffler**, F. Breitling, A. Nesterov-Mueller. Selective Functionalization of Microstructured Surfaces by Laser-Assisted Particle Transfer. *Advanced Functional Materials*, **2016**, 26, 7067–7073

3.8 49. J. Striffler, D. S. Mattes, S. Schillo, B. Muenster, A. Palermo, B. Ridder, A. Welle, V. Trouillet, V. Stadler, G. Markovic, G. Proll, S. Bräse, **F.F. Loeffler**, A. Nesterov-

Mueller, F. Breitling. Replication of Polymer Based Peptide Microarrays by Multi Step Transfer. *ChemNanoMat*, **2016**, 2, 897–903

16.6 50. **F.F. Loeffler***, T. C. Foertsch, R. Popov, D. Althuon, M. Schlageter, M. Sedlmayr, B. Ridder, F.-X. Dang, C. v. Bojničić-Kninski, L. K. Weber, A. Fischer, J. Greifenstein, V. Bykovskaya, I. Buliev, F. R. Bischoff, L. Hahn, M. A. R. Meier, S. Bräse, A. K. Powell, T. S. Balaban, F. Breitling*, A. Nesterov-Mueller*. High-flexibility combinatorial peptide synthesis with laser-based transfer of monomers in solid matrix material. *Nature Communications*, **2016**, 7, 11844

6.7 51. B. Muenster, A. Welle, B. Ridder, D. Althuon, J. Striffler, T. C. Foertsch, L. Hahn, R. Thelen, V. Stadler, A. Nesterov-Mueller*, F. Breitling*, **F.F. Loeffler***. Solid-material-based Coupling Efficiency Analyzed with Time-of-Flight Secondary Ion Mass Spectrometry. *Applied Surface Science*, **2016**, 360, 306–314

pend. 52. A. Nesterov-Mueller*, F. Märkle, L. Hahn, T. Förtsch, S. Schillo, V. Bykovskaya, M. Sedlmayr, L.K. Weber, B. Ridder, M. Soehindrijo, B. Münster, J. Striffler, F. R. Bischoff, F. Breitling, **F.F. Loeffler***. Particle-based microarrays of oligonucleotides and oligopeptides. *MDPI Microarrays*, New and Old Technologies for Generation of Microarrays, **2014**, 3, 245-262

29.4 53. F. Maerkle[#], **F.F. Loeffler**[#] (**shared first author**), S. Schillo, T. Foertsch, B. Muenster, J. Striffler, C. Schirwitz, F. R. Bischoff, F. Breitling, A. Nesterov-Mueller. High-Density Peptide Arrays with Combinatorial Laser Fusing. *Advanced Materials,* **2014**, 26, 3730–3734

29.4 54. C. Schirwitz, **F.F. Loeffler**, T. Felgenhauer, V. Stadler, A. Nesterov-Müller, R. Dahint, F. Breitling, F.R. Bischoff. Purification of High-Complexity Peptide Microarrays by Spatially Resolved Array Transfer to Gold-Coated Membranes. *Advanced Materials*, **2013**, 25, 1598–1602

2.1 55. C. Schirwitz, **F.F. Loeffler***, T. Felgenhauer, V. Stadler, F. Breitling, F.R. Bischoff. Sensing Immune Responses with Customized Peptide Microarrays. *Biointerphases*, **2012**, 7, 47

19.0 56. **F. Loeffler**, C. Schirwitz, J. Wagner, K. Koenig, F. Maerkle, G. Torralba, M. Hausmann, F. R. Bischoff, A. Nesterov-Mueller, F. Breitling. Biomolecule arrays using functional combinatorial particle patterning on microchips. *Advanced Functional Materials*, **2012**, 22, 2503–2508

4.6 57. J. Wagner, K. Koenig, T. Foertsch, **F. Loeffler**, S. Fernandez, T. Felgenhauer, F. Painke, G. Torralba, V. Lindenstruth, V. Stadler, F.R. Bischoff, F. Breitling, M. Hausmann, A. Nesterov-Mueller. Microparticle transfer onto pixel electrodes of 45 µm pitch on HV-CMOS chips – Simulation and experiment. *Sensors and Actuators A: Physical*, **2011**, 172, 533–545

5.2 58. **F. Löffler**, J. Wagner, K. König, F. Märkle, S. Fernandez, C. Schirwitz, G. Torralba, M. Hausmann, V. Lindenstruth, F. R. Bischoff, F. Breitling, A. Nesterov. High-precision combinatorial deposition of micro particle patterns on a microelectronic chip. *Aerosol Science and Technology*, **2011**, 45, 65–74

2.3 59. F. Breitling, **F. Löffler**, C. Schirwitz, Y.C. Cheng, F. Märkle, K. König, T. Felgenhauer, E Dörsam, F.R. Bischoff, A. Nesterov-Müller. Alternative Setups for Automated Peptide Synthesis. *Mini-Reviews in Organic Chemistry*, **2011**, 8, 121–131

1.6 60. J. Wagner, **F. Löffler**, K. König, S. Fernandez, A. Nesterov-Müller, F. Breitling, F. R. Bischoff, V. Stadler, M. Hausmann, V. Lindenstruth. Quality analysis of selective microparticle deposition on electrically programmable surfaces. *Review of Scientific Instruments*, **2010**, 81, 073703

8.4 61. K. König, I. Block, A. Nesterov, G. Torralba, S. Fernandez, T. Felgenhauer, K. Leibe, C. Schirwitz, **F. Löffler**, F. Painke, J. Wagner, U. Trunk, F.R. Bischoff, F. Breitling, V. Stadler, M. Hausmann, V. Lindenstruth. Programmable high voltage CMOS chips for particle-based high-density combinatorial peptide synthesis. *Sensors and Actuators B: Chemical*, **2010**, 147, 418–427

3.4 62. A. Nesterov, **F. Löffler**, Y.-C. Cheng, G. Torralba, K. König, M. Hausmann, V. Lindenstruth, V. Stadler, F. R. Bischoff, F. Breitling. Characterization of triboelectrically charged particles deposited on dielectric surfaces, *Journal of Physics D: Applied Physics,* **2010**, 43, 165301

3.4 63. A. Nesterov, **F. Löffler**, K. König, U. Trunk, K. Leibe, T. Felgenhauer, F.R. Bischoff, F. Breitling, V. Lindenstruth, V. Stadler, M. Hausmann. Measurement of triboelectric charging of moving micro particles by means of an inductive cylindrical probe. *Journal of Physics D: Applied Physics*, **2007**, 40, 6115–6120

1.6 64. A. Nesterov, **F. Löffler**, K. König, U. Trunk, K. Leibe, T. Felgenhauer, V. Stadler, F.R. Bischoff, F. Breitling, V. Lindenstruth, M. Hausmann. Noncontact charge

measurement of moving microparticles contacting dielectric surfaces. *Review of Scientific Instruments*, **2007**, 78, 075111

Book chapters

65. **F.F. Loeffler*** (**shared corresponding author**), J. Pfeil, K. Heiss*. High-density peptide arrays for malaria vaccine development. *Methods in Molecular Biology, Vaccine Design, Methods and Protocols: Volume 1: Vaccines for Human Diseases,* Humana Press **2016**
66. **F.F. Loeffler*** (**shared corresponding author**), Y.-C. Cheng, B. Muenster, J. Striffler, F. C. Liu, F. R. Bischoff, E. Doersam, F. Breitling, A. Nesterov-Mueller*. Printing Peptide Arrays with a Complementary Metal Oxide Semiconductor Chip. Fundamentals and application of new bioproduction systems, *Advances in Biochemical Engineering/Biotechnology* **2013**
67. J. Wagner, **F. Löffler**, T. Förtsch, C. Schirwitz, S. Fernandez, H. Hinkers, H. F. Arlinghaus, F. Painke, K. Koenig, F. R. Bischoff, A. Nesterov-Müller, F. Breitling, M. Hausmann, V. Lindenstruth. Image Processing Quality Analysis for Particle Based Peptide Array Production on a Microchip, in *Advanced Image Acquisition, Processing Techniques and Applications I*, Dimitrios Ventzas (Ed.), ISBN: 978-953-51-0342-4, InTech **2012**
68. **F. Loffler**, Y.-C. Cheng, T. Fortsch, E. Dorsam, R. Bischoff, F. Breitling, A. Nesterov-Muller. Biofunctional Xerography, in *Biotechnology of Biopolymers*, Magdy Elnashar (Ed.), ISBN: 978-953-307-179-4, InTech **2011**
69. A. Nesterov, E. Dörsam, Y.C. Cheng, C. Schirwitz, F. Märkle, **F. Löffler**, K. König, V. Stadler, R. Bischoff, F. Breitling. Peptide Arrays with a Chip. *Methods in Molecular Biology*, *Small Molecule Microarrays*, Vol. 669, Humana Press, **2010**

Proceedings

1. G. Paris, J. Heidepriem, A. Tsouka, M. Mende, S. Eickelmann, **F.F. Loeffler**. Automated laser-assisted synthesis of microarrays for infectious disease research. *Proceedings of SPIE, Microfluidics, BioMEMS, and Medical Microsystems XVII*, **2019**, 108750C
2. L.K. Weber, A. Fischer, T. Schorb, M. Soehindrijo, T.C. Förtsch, C. v. Bojnicic-Kninski, D. Althuon, **F.F. Loeffler**, F. Breitling, J. Hubbuch, A. Nesterov-Müller. Automated Microfluidic System with Optical Setup for the Investigation of Peptide-Antibody Interactions in an Array Format. *Microsystems Technology in Germany* **2016**
3. Y.C. Cheng, **F. Löffler**, K. König, A. Nesterov, E. Dörsam, F. Breitling. Chip Printer. *Proceedings of the 2nd WSEAS International Conference on Nanotechnology*. **2010** ISSN: 1790-5117, 19–22
4. A. Nesterov, **F. Löffler**, M. Hausmann, V. Stadler, F.R. Bischoff, F. Breitling. Manipulation of solid particle aerosols for combinatorial fabrication of molecular libraries. SPIE Proceedings, ILLA'09, ISSN: 1314-068X, Smolyan, October 18-22, **2009**, 332–341

Other publications (invited)

1. **F. Löffler**. Microarrays selbst gebaut. *Transkript*, **2019**, 4
2. **F. Löffler**, F. Breitling, A. Nesterov-Müller. Kombinatorische Chemie im hochdichten Arrayformat. *Biospektrum*, **2016**, 22, 532
3. Introducing Felix Loeffler. *Angewandte Chemie International Edition*, **2023**, 62, e202306960